Joongheon Kim

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Educational Backgrounds

- **University of Southern California (USC)** – Viterbi School of Engineering *Los Angeles, California, USA*
  – Ph.D. (08/2009–08/2014) in **Computer Science**, Thomas Lord Department of Computer Science
  Advisor: Prof. Andreas F. Molisch (Communications, Information, Learning & Quantum (CLIQ) Group)
  – M.S. (05/2014) in **Computer Science** with specialization in **High Performance Computing and Simulations**
  – M.S. (05/2012) in **Electrical Engineering**
- **Korea University** – College of Informatics *Seoul, Republic of Korea*
  – M.S. (03/2004–02/2006) in **Computer Science and Engineering**
  – B.S. (03/1999–02/2004) in **Computer Science and Engineering**

Professional Affiliations

Full-Time

- **Korea University**, *Seoul, Republic of Korea*
  Professor (09/2019–Present), Department of Electrical and Computer Engineering
  Adjunct Professor (03/2023–Present), Department of Communications Engineering (Samsung Electronics)
  Adjunct Professor (03/2021–Present), Department of Semiconductor Engineering (SK Hynix)
  Adjunct Professor (11/2022–Present), Department of Future Science and Technology Business (Graduate School)
  Affiliated Professor (03/2025–Present), Department of Defense Convergence Technology (LIG Nex1, Graduate School)
- **Chung-Ang University**, *Seoul, Republic of Korea*
  Assistant Professor (03/2016–08/2019), School of Computer Science and Engineering
- **Intel Corporation**, *Santa Clara (Silicon Valley), California, USA*
  Systems Engineer (09/2013–02/2016), Platform Engineering Group
- **University of Southern California – Viterbi School of Engineering**, *Los Angeles, California, USA*
  Ph.D. Student in Computer Science (08/2009–08/2014), Thomas Lord Department of Computer Science
  Annenberg Graduate Fellow (2009), 4-Year Full Scholarship for the Ph.D. Program in Computer Science
  Research Assistant (01/2011–08/2014), Communications, Information, Learning & Quantum (CLIQ) Group
  Teaching Assistant (01/2012–05/2013), Computer Science and Electrical Engineering Major Courses
- **InterDigital**, *San Diego, California, USA*
  Intern (05/2012–08/2012), Wireless Systems Evolution Department
- **LG Electronics – CTO Office**, *Seoul, Republic of Korea*
  Research Engineer (01/2006–08/2009), Multimedia Research Lab, Seocho R&D Campus

Joint Appointments for Quantum Research Collaboration

- **Seoul National University Hospital**, *Seoul, Republic of Korea*
  Visiting Professor (10/2025–Present), Healthcare AI Research Institute

Services for National R&D

- **Presidential Council on National Artificial Intelligence Strategy**, **Advisory Committee (12/2025–Present)**
- **National Research Foundation of Korea**, **Review Board (11/2024–10/2026)**

University R&D/Administrative Positions

R&D Leadership Positions

- **Korea University**, *Seoul, Republic of Korea*
  Director (07/2024–12/2031), Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center
  Principal Investigator, Software Star-Lab (07/2024–12/2031), Quantum AI Empowered Second-Life Platform Technology

Administrative Positions

- **Korea University**, *Seoul, Republic of Korea*
  Associate Dean (08/2026), University College
  Vice Department Chair (01/2025–08/2025), Academic Affairs, School of Electrical Engineering
  Deputy Vice President (02/2022–08/2024), Office of Academic Affairs
  Dean (06/2021–08/2023), Center for Teaching and Learning (CTL)

Committee Members

- **Korea University**, *Seoul, Republic of Korea*
  Member (12/2023–Present), Council of the Interdisciplinary Program in Quantum Information Science

Awards and Honors

Research and Academic Excellence (International)

- **Top Cited Article in ETRI Journal (Wiley), among work published in 2024** 2025
- **Finalist, AAAI Student Abstract and Poster Session – Oral Presentation** 2026 (Top 35), 2023 (Top 25)
- **IEEE ICTC Best Paper Award** – *IEEE Communications Society* 2022
- **Spotlight, Oral Presentation** – *ICML Workshop on Dynamic Neural Networks (2022)* 2022
- **IEEE MMTC Best Journal Paper Award** – *IEEE Communications Society* 2021
- **IEEE ICOIN Best Paper Award** – *IEEE Computer Society* 2021
- **IEEE MMTC Outstanding Young Researcher Award** – *IEEE Communications Society* 2020
- **IEEE Systems Journal Best Paper Award** – *IEEE Systems Council* 2020
- **Next Generation and Standards (NGS) Division Recognition Award** – *Intel Corporation* Q1/2015
- **Annenberg Graduate Fellowship Award** – *University of Southern California* 2009–2013
  Awarded with Ph.D. Admission: 4 Year Full Scholarship (Tuition Waiver and $120,000 Stipend ($30,000/year for 4 years))
- **IEEE Seoul Section Student Paper Contest** Gold (2019), Silver (2025), Bronze (2025, 2024, 2024, 2023, 2020)
- **IEEE VTS Seoul Chapter Award** – *IEEE Vehicular Technology Society* 2023, 2022, 2022, 2021, 2021, 2019

Research and Academic Excellence (Korea Regional, *since 2016*)

- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications)** – *KICS* 2025
- **Best Paper Award, The Journal of KICS** – *KICS* 2024
- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications)** – *KICS* 2023
- **Haedong Young Scholar Award** – *KICS* and *Haedong Foundation* 2018
  For recognizing a researcher under the age of 40 who has made outstanding contributions to IT R&D

Korea University

- **Granite Tower Best Research Award (Top 3%)**, Totally, 2 times
  – 2025 (Top 3%, top 53 professors at Korea University in 2025) 05/2026
  – 2024 (Top 3%, top 54 professors at Korea University in 2024) 05/2025
- **Best Research Achievement Award** –*Korea University, School of Electrical Engineering* 2024
- **Insung Research Grant Award** – *Korea University* 01/2023
  For recognizing professors in top 5% research excellence during the first 3 years at Korea University
- **Granite Tower Best Teaching Award (Top 5%)**, Totally, 7 times
  – Probability and Random Process KECE209-02 (2025)
  – Software Programming Basics GECT002-01 (2024), GECT002-06 (2025)
  – Computer Language and Laboratory EGRN151-07 (2019), EGRN151-06 (2021)
  – Introduction to Computers SEMI103 (2021)
  – Future Mobility Technology GEQR075 (2022)
- **Best Teaching Award (Top 20%)**, Totally, 7 times
  – Probability and Random Process KECE209-02 (2021), KECE209-02 (2022)
  – Software Programming Basics GECT002-02 (2024), GECT002-08 (2024), GECT002-09 (2025)
  – Computer Language and Laboratory EGRN151-09 (2020)
  – Object-Oriented Programming SEMI104 (2021)

Academic and University Services (International)

- **Certificate of Appreciation** – *IEEE Future Networks, AI/ML Working Group* 08/2025
- **Certificate of Appreciation** – *IEEE/IFIP WiOpt (2024)* 10/2024
- **Best Editor Award** – *ICT Express (Elsevier)* 07/2023
- **2022 Best Chapter Award, IEEE Vehicular Technology Society Chapter**, Awarded as a Treasure 2022
- **Appreciation Recognition** – *Daegu Gyeongbuk Institute of Science and Technology (DGIST)* 2021
- **Certificate of Appreciation** – *Department of Computer Science, University of Southern California* 2010

Academic and University Services (Korea Regional)

- **Outstanding Contribution Award**
  – The Korean Institute of Communication Sciences and Information Sciences (KICS) 2025, 2024, 2021, 2019
  – Korean Institute of Information Scientists and Engineers (KIISE) Information Network Society 2023, 2022
  – Open Standards and ICT Association (OSIA) 2025, 2021
  – The Institute of Electronics and Information Engineers (IEIE) 2025
- **Commemorative Plaque**, *KICS Excellent Lecture* (Title: Quantum Machine Learning and Software Technologies) 2025
- **Best Group Award**, *KICS Research Group in Quantum Communications and Quantum Computing* (Group Leader) 2025
- **Appreciation Recognition** – *Daegu Gyeongbuk Institute of Science and Technology (DGIST)* 2021

Student Awards (International)

- **Mitacs Globalink Research Award - for research in Canada**, Awarded to *Seok Bin Son* 2026
  Project Title) Quantum Federated Learning in Distributed Systems: Challenges and Future Directions
  Advisor) Prof. Trung Q. Duong (Memorial University of Newfoundland)

- **Mitacs Globalink Research Award - for research in Canada**, Awarded to *Seungcheol Oh* 2026
  Project Title) Over the Air Quantum Federated Learning
  Advisor) Prof. Octavia Dobre (Memorial University of Newfoundland)
- **IEEE TCSC Award for Excellence in Scalable Computing** (IEEE Computer Society), Awarded to *Soohyun Park* 2025
- **Young Scientist Award** (ICSANE 2025), Awarded to *Minji Lee* 2025
- **IEEE Vehicular Technology Society Student Scholarship Award (2025)**, Awarded to *Gyu Seon Kim* 2025
- **IEEE ICDCS Best Runner-Up Poster Paper Award (2025)**, Awarded to *Seok Bin Son*, *Soohyun Park* 2025

R&D Projects

University-Wide/Center Projects

- **Net-Zero CAFE Research Center** (Korea University) **[Center Director, PI]**
  IT Research Center (ITRC) *by* Ministry of Science and ICT (7,500,000,000 KRW) (07/2024–12/2031)
- **Intelligent 6G Wireless Access System Research Center** (Korea University)
  6G AI Research Center *by* Ministry of Science and ICT (290,000,000 KRW) (04/2021–12/2025)
- **K-Starlink: Dynamic Reconfigurable and Intelligent Space-Terrestrial Networks** (Ajou University)
  Basic Research Lab (BRL) *by* Ministry of Science and ICT (251,000,000 KRW) (06/2021–05/2024)
- **Nano UAV Intelligence Systems Research Lab** (Kwangwoon University)
  ADD Military Special Research Center *by* Agency for Defense Development (130,000,000 KRW) (10/2020–08/2023)
- **5G/Unmanned Vehicle Research Center (5G/UV-RC)** (Hanyang University)
  IT Research Center (ITRC) *by* Ministry of Science and ICT (55,709,000 KRW) (06/2020–12/2022)
- **Human Resource Development for the Biomedical Unstructured Big Data Analysis** (SNU Hospital)
  IT Research textrm (ITRC) *by* Ministry of Science and ICT (08/2018–12/2021)
- **Novel Data Science Driven Framework for Efficient Network Design** (Chung-Ang University)
  Basic Research Lab (BRL) *by* Ministry of Science and ICT (150,000,000 KRW) (06/2017–05/2020)

Consortium Projects

- **6GARROW: 6G AI-Native Integrated RAN-Core Networks**
  (Korea-Europe International Collaboration Research for Digital Core Technologies) (256,740,000 KRW) (09/2024–08/2027)
  Institute for ICT Planning and Evaluation (IITP) *by* Ministry of Science and ICT
- **AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development**
  (Human-Centric AI Core Technology Development Research) (537,222,852 KRW) (04/2022–12/2026)
  Institute for ICT Planning and Evaluation (IITP) *by* Ministry of Science and ICT
- **Development of Integrated Development Framework that supports Automatic Neural Network Generation and Deployment optimized for Runtime Environment**
  (Core Technology Development Project for Software Computing Industry) (291,073,362 KRW) (04/2021–12/2025)
  Institute for ICT Planning and Evaluation (IITP) *by* Ministry of Science and ICT
- **Integrated Perception Technology Developments for Public Safety Platforms**
  (Complex Cognitive Technology Development Project) (400,000,000 KRW) (06/2019–05/2023)
  National Research Foundation of Korea *by* Ministry of Science and ICT
- **Development of Privacy-Reinforcing Distributed Transfer-Iterative Learning Algorithm**
  (Healthcare and Medical Technologies Research and Development (R&D) Project) (149,558,670 KRW) (07/2019–12/2021)
  Ministry of Health and Welfare
- **Virtual Presence in Moving Objects through 5G (PriMO-5G)**
  (Korea-Europe International Collaboration Research) (246,464,000 KRW) (06/2018–06/2021)
  Institute for ICT Planning and Evaluation (IITP) *by* Ministry of Science and ICT
- **Distributed Secure Platform for Scalable Clinical OMOP CDM Models**
  (Medical Data Protection Research and Development (R&D) Project) (90,000,000 KRW) (04/2019–12/2020)
  Ministry of Health and Welfare

Industry Projects

- **Advancement Technology Development for Torpedo Deception Strategies in Submarines**
  LIG Defense and Aerospace (LIG D&A) (700,000,000 KRW) (11/2022–11/2026)
- **Advancement Technology Development for Submarine Target Identification and Engagement Support Intelligence**
  LIG Defense and Aerospace (LIG D&A) (300,000,000 KRW) (11/2022–11/2026)
- **mmWave-based Liquid Classification in Next-Gen Robotic Vacuum Cleaners**
  LG Electronics, Robotics Research Lab (HS Future Technology Project) (30,000,000 KRW) (06/2026–09/2026)
- **Mapping between Real World and VR for End-Edged Cloud Real-Time VR Servers**
  Samsung Electronics (357,500,000 KRW) (09/2020–11/2025)
- **Research on Learning-based Swarm Mission Planning Algorithms**
  LIG Defense and Aerospace (LIG D&A) (110,000,000 KRW) (03/2024–02/2025)
- **Quantum Machine Learning-based Objection Detection for Point Cloud and its Acceleration**
  Hyundai Motors Group (110,000,000 KRW) (12/2022–04/2024)
- **Routing Algorithms for LEO Satellite Networks**
  Solvit System (27,500,000 KRW) (12/2022–08/2023)

- **Optimal Positioning Algorithms for Wide-Area Relaying Networks**
  Solvit System (22,000,000 KRW) (12/2022–08/2023)
- **Distributed Learning Algorithms to Build AI Models with Multi-Center Clinical Data**
  Cipherome (12,000,000 KRW) (11/2022–02/2023)
- **Cellular/Wi-Fi Handover Technology Development**
  LG Electronics CTO Division, Smart Mobility Lab, Advanced R&D Center (88,000,000 KRW) (02/2022–12/2022)
- **Research Trends in Digital Twin Applications to Autonomous Driving**
  Hyundai NGV (1,000,000 KRW) (03/2022–04/2022)
- **Distributed Learning System Design and Implementation for Clinical Applications**
  Cipherome (15,000,000 KRW) (02/2022–03/2022)
- **Super-Resolution Performance Optimization in Mobile Platforms**
  Samsung SDS (15,000,000 KRW) (05/2020–08/2020)
- **Deep Learning Algorithms for mVOC Concentration Analysis**
  Samsung Electronics, C-Lab (12,000,000 KRW) (03/2020–06/2020)
- **Visual Recognition Software Implementation using Deep Learning Tools**
  Hyundai Motors Group (48,189,170 KRW) (05/2019–11/2019)
- **A Priori Techniques Research for Efficient Multi-Edge Computing**
  Samsung Electronics, Software Center (80,000,000 KRW) (06/2017–12/2017)

Government Projects

- **Quantum AI Empowered Second-Life Platform Technology**
  (Software Star-Lab, Software Technology Advanced Research) (2,200,000,000 KRW) (07/2024–12/2031)
  Institute for ICT Planning and Evaluation (IITP) *by* Ministry of Science and ICT
- **Quantum-Empowered Spatio-Temporal Multi-Scale Digital Twin System**
  (Mid-Career Researcher Support Program) (750,000,000 KRW) (03/2025–02/2028)
  National Research Foundation of Korea *by* Ministry of Science and ICT
- **SEDNA: Satellite-aided Entanglement Distribution for Networked Applications** (with *Prof. Symeon Chatzinotas*)
  (Korea-Luxembourg Quantum Science and Technology Programme) (50,000,000 KRW) (10/2026–09/2027)
  National Research Foundation of Korea *by* Ministry of Science and ICT
- **Research Planning for RL-based Scheduling in Autonomous Satellite Task Allocation**
  (Private Sector-Linked Public Research Outcome Dissemination Support Program) (35,000,000 KRW) (05/2026–10/2026)
  Korea Industrial Technology Association (KOITA)
- **Quantum Hyper-Driving:**
  Quantum-Inspired Hyper-Connected and Hyper-Sensing Autonomous Mobility Technologies
  (Mid-Career Researcher Support Program) (571,836,753 KRW) (03/2022–02/2025)
  National Research Foundation of Korea *by* Ministry of Science and ICT
- **Korea-Japan Joint Seminar Project for Generative and Multi-Modal AI Technologies**
  (International Research Collaboration Program) (50,000,000 KRW) (10/2023–09/2024)
  National Research Foundation of Korea *by* Ministry of Science and ICT
- **Development of Quantum Deep Reinforcement Learning Algorithm using QAOA**
  (Quantum Computing Technology Development Project) (503,361,794 KRW) (10/2019–04/2022)
  National Research Foundation of Korea *by* Ministry of Science and ICT
- **mmWave Radar and Deep Reinforcement Learning based Optimal Policy Autonomous Driving**
  (Mid/Early-Career Researcher Support Program) (282,124,926 KRW) (06/2019–02/2022)
  National Research Foundation of Korea *by* Ministry of Science and ICT
- **mmWave High-Speed Networking Platform Design for Next-Generation Convergence Services**
  (Early-Career Researcher Support Program, also selected as Initial Innovation Lab) (210,000,000 KRW) (06/2016–05/2019)
  National Research Foundation of Korea *by* Ministry of Science and ICT
- **Feasibility Study of 60 GHz IEEE 802.11ad for Virtual Reality (VR) Platforms**
  (Broadcasting and Telecommunications Industry Technology Development Project) (33,333,000 KRW) (04/2017–12/2017)
  Institute for ICT Planning and Evaluation (IITP) *by* Ministry of Science and ICT

Government-Funded Research Institute Projects

- **Research on Trainability and Learnability in Quantum Machine Learning Models**
  ETRI-Affiliated Institute (140,000,000 KRW) (03/2026–10/2027)
- **LEO Satellite Routing Research using Large Language Model and Reinforcement Learning**
  Electronics and Telecommunications Research Institute (ETRI) (40,000,000 KRW) (05/2025–11/2025)
- **Discovering Quantum-Advantage Cases for Industry Adoption**
  Korea Testing Laboratory (KTL) (16,500,000 KRW) (09/2025–11/2025)
- **Research on Generative Quantum Machine Learning Models**
  ETRI-Affiliated Institute (80,000,000 KRW) (04/2025–10/2025)
- **Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks**
  Electronics and Telecommunications Research Institute (ETRI) (40,000,000 KRW) (05/2024–11/2024)
- **NOMA-based Resource Allocation Research in Space-Air-Ground Integrated Networks**
  Electronics and Telecommunications Research Institute (ETRI) (20,900,000 KRW) (09/2023–11/2023)

- **Autonomous Intelligent COA Search Methods for Cyber-Attacks**
  Agency for Defense Development (ADD) (104,039,000 KRW) (12/2021–11/2022)
- **Research on Intelligent Agent-based CPS Security and Reliability**
  Telecommunications Technology Association (TTA) (48,000,000 KRW) (04/2021–11/2021)
- **Multi-GPU based Automotive HPC Platform Development**
  Electronics and Telecommunications Research Institute (ETRI) (20,000,000 KRW) (04/2020–10/2020)
- **Cooperative Deep Reinforcement Learning for Online Game Multi-Agents**
  Electronics and Telecommunications Research Institute (ETRI) (28,000,000 KRW) (04/2020–08/2020)
- **Verification Testbed Implementation for Privacy-Preserving Trust Data Generation**
  Electronics and Telecommunications Research Institute (ETRI) (44,000,000 KRW) (10/2019–11/2019)
- **Measurement and Analysis of Multi-Task GPU Scheduling Delays**
  Electronics and Telecommunications Research Institute (ETRI) (40,000,000 KRW) (05/2019–10/2019)
- **Probabilistic Decision Making and Econometric Methods for Micro-Grid**
  Korea Electric Power Corporation (KEPCO) Research Institute (143,128,000 KRW) (05/2017–04/2019)
- **GPU Scheduling Performance Analysis under Queueing Delay Considerations**
  Electronics and Telecommunications Research Institute (ETRI) (40,000,000 KRW) (05/2018–10/2018)
- **Improving Massive Deep Learning Training via Computation and Communication Acceleration**
  Electronics and Telecommunications Research Institute (ETRI) (30,000,000 KRW) (04/2018–10/2018)
- **Parsing Techniques for Artificial Neural Network (ANN) Data Processing**
  Electronics and Telecommunications Research Institute (ETRI) (40,000,000 KRW) (09/2017–11/2017)

University of Southern California, Selected Projects

- **Video Aware Wireless Networks (VAWN) Research Program** — Intel Labs, Verizon Wireless, Cisco Systems
- **60 GHz Real-Time Wireless Video Broadcasting** — Disney Research Zürich
- **Annenberg Graduate Fellowship Award** — University of Southern California

Korea University (KU), Selected Projects

- **Towards Agentic AI-Enabled Edge IoT Systems for Proactive Physical World Sensing and Control** (08/2025–07/2026)
  PKU-KU Joint Research Grant (Collaboration with Prof. Kaigui Bian at Peking University)
- **AI Teaching Assistant Research for LLM-based Large-Scale Education** (10/2024–02/2025) — KU University College
- **Autonomous Mobility Control using Quantum Deep Learning** (03/2023–02/2024) — KU Insung Research Grant
- **Mobile Access Algorithm Design using Economic Theory and AI** (03/2020–02/2021) — KU Future Research Grant

Selected Publications

- **12,683+ Citations** (H-index: 55+, i10-index: 260+), obtained from Google Scholar Profile (as of November 1, 2026)
- **Journals and Magazines:** **183** publications; among them, **141** IEEE/Nature/APS publications

Books

- *Fundamentals of 6G Communications and Networking*, Springer (2024) *(Editors: X. Lin, J. Zhang, Y. Liu, J. Kim)*

Selected Papers

■ Top-Tier Conference Publications

[ACCV'26] Personalized Quantum Federated Visual Learning via Raced Architecture Search and Ensemble Calibration
ACCV 2026 (Osaka, Japan, December 2026) *(S. B. Son, S. Park, J. Kim, W. Lee)*

[CIKM'26] Strong Lottery Ticket Hypothesis for Quantum Neural Networks
ACM CIKM 2026 (Rome, Italy, November 2026) *(H. Kwon, H. K. Lee, J. Kim, S. W. Yoon)*

[CIKM'26] Beneath the Noise: Uncovering and Distilling the Encoding Bottleneck of Quantum Models
ACM CIKM 2026 (Rome, Italy, November 2026) *(S. Lee, C. Kim, S. Son, J. Kim, H. In)*

[ICML'26] Quantum Robust Inner Minimization for Reinforcement Learning with Quadratic Speed-Up in Query Complexity
ICML 2026 (Seoul, Korea, July 2026) *(H. K. Lee, J. Kim, S. W. Yoon)*

[ACL'26] Let LLM Tutors Ask First: Proactive LLM-Based Tutoring at Scale in a 1,500-Student Online Classroom
ACL 2026 (San Diego, CA, USA, July 2026) *(J. Lee, G. Youn, S. Lee, C. Im, J. Kim, C. Yoo)*

[CIKM'25] Quantum-Amplitude Embedded Adaptation for Parameter-Efficient Fine-Tuning in Large Language Models
ACM CIKM 2025 (Seoul, Korea, November 2025) *(E. J. Roh, J. Kim)*

[CIKM'25] Filtered One-Shot Training for Quantum Architecture Search
ACM CIKM 2025 (Seoul, Korea, November 2025) *(S. B. Son, S. Y.-C. Chen, J. Kim, S. Park)*

[CIKM'25] LLM-based Interactive Coding Education via Predictive Query Management and Student-Centered Fine-Tuning: Design and Implementation with 1500-Student Class Data
ACM CIKM 2025 (Seoul, Korea, November 2025) *(G. Youn, J. Lee, J. Kim, C. Yoo)*

[IPDPS'25] AQUA: Hardware-Agnostic Qubit Allocation for Quantum Multi-Programming
IEEE IPDPS 2025 (Milan, Italy, June 2025) *(X. Piao, J. Shim, J. Kim, J. Kim)*

[WiOpt'25] Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach
IEEE/IFIP WiOpt 2025 (Linköping, Sweden, May 2025) *(G. S. Kim, J. Chung, T. Q. Duong, S. Park, J. Kim)*

[CIKM'24] Hands-On Introduction to Quantum Machine Learning
ACM CIKM 2024, *Tutorial* (Boise, ID, USA, October 2024) *(S. Y.-C. Chen, J. Kim)*

[WiOpt'24] Advanced Taxiing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management
IEEE/IFIP WiOpt 2024 (Seoul, Korea, October 2024) *(S. Lee, G. S. Kim, S. Park, J. Kim)*

[CIKM'23] Quantum Split Learning for Privacy-Preserving Information Management
ACM CIKM 2023 (Birmingham, United Kingdom, October 2023) *(S. Park, H. Baek, J. Kim)*

[CIKM'23] Logarithmic Dimension Reduction for Quantum Neural Networks
ACM CIKM 2023 (Birmingham, United Kingdom, October 2023) *(H. Baek, S. Park, J. Kim)*

[AAAI'23] Quantum Multi-Agent Meta Reinforcement Learning
AAAI 2023 (Washington DC, USA, February 2023) *(W. Yun, J. Park, J. Kim)*

[CIKM'22] Hierarchical Reinforcement Learning using Gaussian Random Trajectory Generation in Autonomous Furniture Assembly
ACM CIKM 2022 (Atlanta, GA, USA, October 2022) *(W. Yun, D. Mohaisen, S. Jung, J.-K. Kim, J. Kim)*

[WiOpt'22] Cooperative Video Quality Adaptation for Delay-Sensitive Dynamic Streaming using Adaptive Super-Resolution
IEEE/IFIP WiOpt 2022 (Torino, Italy, September 2022) *(M. Choi, W. Yun, J. Kim)*

[INFOCOM'22] Joint Superposition Coding and Training for Federated Learning over Multi-Width Neural Networks
IEEE INFOCOM 2022 (Virtual, May 2022) *(H. Baek, W. Yun, Y. Kwak, S. Jung, M. Ji, M. Bennis, J. Park, J. Kim)*

[ICDCS'20] Understanding the Potential Risks of Sharing Elevation Information on Fitness Applications
IEEE ICDCS 2020 (Virtual, November/December 2020) *(Ü. Meteriz, N. F. Yildiran, J. Kim, D. Mohaisen)*

[IJCAI'19] Randomized Adversarial Imitation Learning for Autonomous Driving
IJCAI 2019 (Macau, China, August 2019) *(M. Shin, J. Kim)*

[ICDCS'18] ShmCaffe: A Distributed Deep Learning Platform with Shared Memory Buffer for HPC Architecture
IEEE ICDCS 2018 (Vienna, Austria, July 2018) *(S. Ahn, J. Kim, E. Lim, W. Choi, A. Mohaisen, S. Kang)*

[ACM-MM'17] REQUEST: Seamless Dynamic Adaptive Streaming over HTTP for Multi-Homed Smartphone under Resource Constraints
ACM Multimedia 2017 (Mountain View, CA, USA, October 2017) *(J. Koo, J. Yi, J. Kim, M. Hoque, S. Choi)*

[MobiSys'10] Energy-Efficient Rate-Adaptive GPS-based Positioning for Smartphones
ACM MobiSys 2010 (San Francisco, CA, USA, June 2010) *(J. Paek, J. Kim, R. Govindan)*

■ Magazines and Journals

◄ Accepted ►

[CIM.*accepted*] **Rethinking Privacy-Preserving Distributed Learning in Quantum Hilbert Spaces**, ***IEEE Computational Intelligence Magazine***, v(n):ppp–ppp *(S. B. Son, J. Kim, S. Park)*

[CM.*accepted*] **Living on the Quantum Edge: Agentic Quantum AI for Mobile Networks**, ***IEEE Communications Magazine***, v(n):ppp–ppp *(H. Ahn, H. Lee, S. Jung, M. Bennis, S. Park, J. Kim)*

[NM.*accepted*] **Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks**, ***IEEE Network***, v(n):ppp–ppp *(H. Lee, G. S. Kim, E. J. Roh, S. Jung, S. Park, T. Q. Duong, J. Kim)*

[IOTJ.*accepted*] Adaptive 3D Waypoint Navigation for Autonomous Underwater Vehicles Using Contextual Scene Transformer Policy Optimization, *IEEE Internet of Things Journal*, 14(n):ppp–ppp *(E. J. Roh, J. Lee, I. Song, J. Kwon, S. Kwon, Y. Kim, J. Kim)*

[ICTE.*accepted*] A Survey on AI Methods for Energy-Efficient Cell-Free MIMO Networks, *ICT Express*, v(n):ppp–ppp *(M. Lee, S. Oh, J. Kim, S. Park)*

◄ 2026 ►

[IOTJ'26.12] Multi-Agent Transformer for Computation-Efficient User Scheduling in Cell-Free IoT Networks, *IEEE Internet of Things Journal*, 13(24):ppp–ppp *(M. Lee, S. Oh, S. Park, S. Jung, J. Kim)*

[IOTJ'26.12] Carbon-Aware Neural Resource Allocation for Sustainable Integrated NTN-TN Networks, *IEEE Internet of Things Journal*, 13(23):ppp–ppp *(J. Jang, J. Kim, J. Kim, S. Jung)*

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◄ 2018 ►

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Journal of Real-Time Image Processing, 12(2):465-471 *(J. Kim, E. Ryu)*

◄ 2007–2015 ►

[TII'15.12] Energy-Efficient Dynamic Packet Downloading for Medical IoT Platforms, *IEEE Transactions on Industrial Informatics*, 11(6):1653–1659 *(J. Kim)*

[TSMC'15.11] Stochastic Decision Making for Adaptive Crowdsourcing in Medical Big-Data Platforms, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 45(11):1471–1476 *(J. Kim, W. Lee)*

[MTAP'15.10] Interference Impacts on 60 GHz Real-Time Online Video Streaming in Wireless Smart TV Platforms, *Multimedia Tools and Applications*, 74(19):8613-8629 *(J. Kim, S.-N. Hong)*

[TS'15.05] Dynamic Two-Stage Beam Training for Energy-Efficient Millimeter-Wave 5G Cellular Systems, *Telecommunication Systems*, 59(1):111-122 *(J. Kim, S.-N. Hong)*

[CEE'15.04] Adaptive Buffer Control for Distributed Autonomous Robust Routing in Mobile Surveillance Robots, *Computers and Electrical Engineering*, 43:306-316 *(J. Kim, S.-N. Hong)*

[JCN'14.10] Fast Millimeter-Wave Beam Training with Receive Beamforming, *IEEE/KICS Journal of Communications and Networks*, 16(5):512–522 *(J. Kim, A. F. Molisch)*

[EL'14.10] Quality of Video Streaming in 38 GHz Millimetre-Wave Heterogeneous Cellular Networks, *IET Electronics Letters*, 50(21):1526-1528 *(J. Kim, E. Ryu)*

[CL'14.09] Joint Coding and Stochastic Data Transmission for Uplink Cloud Radio Access Networks, *IEEE Communications Letters*, 18(9):1619–1622 *(S.-N. Hong, J. Kim)*

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Academic Activities (Selected: Quantum AI and Quantum Computing)

Invited Talks at Universities (Selected)

- Quantum Neural Networks: Algorithms and Applications
  Keio University (Tokyo, Japan, 12/2025)
- Quantum Reinforcement Learning: From Foundations to Emerging Applications
  Ulsan National Institute of Science and Technology (UNIST) (Ulsan, Korea, 12/2025)
- Quantum Reinforcement Learning: Algorithms and Applications
  Indiana University, Bloomington (Bloomington, Indiana, USA, 10/2025)
- Quantum Neural Networks: Theory, SW, and Applications
  Gwangju Institute of Science and Technology (GIST) (Gwangju, Korea, 05/2025)
- Quantum Neural Networks: Theory, SW, and Applications
  Seoul National University Hospital (Seoul, Korea, 03/2025)
- Quantum Neural Networks: Theory, SW, and Applications
  Sungkyunkwan University (Suwon, Korea, 01/2025)
- Quantum Neural Networks: Theory, SW, and Applications
  Yonsei University (Seoul, Korea, 09/2023)

Journal Special Issues (Guest Editorials)

- Quantum Technologies for Communication Systems (2025), *IEEE/KICS Journal of Communications and Networks*
- Advances in Quantum Networks: Enabling Computing, Security, and AI Innovations (2025), *Discover Networks (Springer)*
- Quantum Algorithms, Optimization, and A.I. (2025), *The Journal of Supercomputing (Springer)*

Organizing Committee Contributions for International Conferences, Symposiums, and Workshops

- Quantum Neural and Reinforcement Learning for Adaptive Sequential Intelligence (ID 47)
  IEEE QAI 2026 (Nottingham, United Kingdom, 12/2026)
- Workshop on Quantum Computing and Reinforcement Learning (2025)
  IEEE Quantum Week (QCE) 2026 (Toronto, Ontario, Canada, 09/2026)
- Quantum Technologies for Earth Observation (2026)
  IEEE IGARSS 2026 (Washington DC, USA, 08/2026)
- Special Session: Quantum Machine Learning Algorithms and Applications (2026)
  IEEE WCCI 2026 (Maastricht, The Netherlands, 06/2026)
- Workshop on Quantum Machine Learning in Signal Processing and Artificial Intelligence (2026)
  IEEE ICASSP 2026 (Barcelona, Spain, 04/2026)

- Workshop: Quantum Computing - Artificial Intelligence (QC-AI) for Industry Applications and Scientific Discovery (2026)
  IEEE QCNC 2026 (Kobe, Japan, 04/2026)
- Quantum Computing for Classical Computational Challenges in HPC (2026)
  IEEE QCNC 2026 (Kobe, Japan, 04/2026)
- Workshop on Quantum Machine Learning (2026)
  IEEE ICOIN 2026 (Hanoi, Vietnam, 01/2026)
- Workshop on Quantum Computing for Communications and Learning (2025)
  IEEE GLOBECOM 2025 (Taipei, Taiwan, 12/2025)
- Workshop on Quantum Computing and Reinforcement Learning (2025)
  IEEE Quantum Week (QCE) 2025 (Albuquerque, New Mexico, USA, 09/2025)
- Workshop on Quantum Machine Learning in Signal Processing and Artificial Intelligence (2025)
  IEEE ICASSP 2025 (Hyderabad, India, 04/2025)
- Workshop on Quantum Computing and Reinforcement Learning (2024)
  IEEE Quantum Week (QCE) 2024 (Montreal, Quebec, Canada, 09/2024)
- Workshop on Quantum Algorithms, Optimization, and Artificial Intelligence (2024)
  IJCAI 2024 (Jeju, Korea, 08/2024)

Program Committee Contributions for International Conferences, Symposiums, and Workshops

- Workshop on Secure and Trustworthy Quantum Machine Learning (2026)
  NeurIPS 2026 (Atlanta, Georgia, USA, 12/2026)
- Workshop on Quantum Machine Learning (2026)
  IEEE ICOIN 2026 (Hanoi, Vietnam, 01/2026)
- Workshop on Quantum Computing for Communications and Learning (2025)
  IEEE GLOBECOM 2025 (Taipei, Taiwan, 12/2025)
- Workshop on Quantum Data and Machine Learning: Systems, Theory and Hardware (2025)
  IEEE ICDE 2025 (Hong Kong ,China, 05/2025)
- Workshop on Quantum Computing and Artificial Intelligence (2024)
  AAAI 2024 (Philadelphia, Pennsylvania, USA, 02/2024)

International Tutorials and Speeches

- Hands-On Quantum AI for Sequential Decision-Making: Variational Quantum Models, Quantum Reinforcement Learning, and Self-Programming Quantum Circuits (ID 46)
  IEEE QAI 2026 Tutorial (Nottingham, United Kingdom, 12/2026)
- Introduction to Distributed Quantum Artificial Intelligence: Neural Networks, Reinforcement Learning, and Multi-Agent Systems
  IEEE WCCI 2026 Tutorial (Maastricht, The Netherlands, 06/2026)
- Quantum-Enhanced Learning for Future Communication Networks
  IEEE GLOBECOM 2025 Tutorial (Taipei, Taiwan, 12/2025)
- Quantum Reinforcement Learning: From Foundations to Emerging Applications
  IEEE QAI 2025 Tutorial (Naples, Italy, 11/2025)
- Quantum Machine Learning: Algorithms and ICT Convergence Applications
  IEEE ICTC 2025 Invited Talk (Jeju, Korea, 10/2025)
- Hands-On Introduction to Quantum Machine Learning
  IEEE QCE 2025 Tutorial (Albuquerque, New Mexico, USA, 09/2025)
- Quantum Reinforcement Learning for Computer Networks and Autonomous Mobility: Modern Trends and Future Directions
  IEEE VTS APWCS 2025 Invited Speech (Tokyo, Japan, 08/2025)
- Quantum Reinforcement Learning: Algorithms and Applications
  IEEE Future Networks Webinar (Future Networks AI/ML and QIT Working Groups) (Webinar, 08/2025)
- Introduction to Quantum Machine Intelligence
  IEEE SSCI 2025 Tutorial (Trondheim, Norway, 03/2025)
- Hands-On Introduction to Quantum Machine Learning
  ACM CIKM 2024 Tutorial (Boise, Idaho, USA, 10/2024)
- Quantum Reinforcement Learning: Concepts, Models, and Applications
  ICIT 2024 Keynote Speech (Danang, Vietnam, 09/2024)
- A Paradigm Shift in Future Networks with Quantum Deep Learning
  IEEE ICUFN 2022 Tutorial (Barcelona, Spain, 07/2022)

Academic Activities and Research Supervision

Research Supervision

■ Ph.D. Alumni

1. **Soohyun Park** (03/2019–08/2023 (PhD), 09/2023–02/2024 (Postdoc)), **Sookmyung Women's University (Professor)**
2. **Hankyul Baek** (03/2021–02/2024 (PhD), 03/2024–03/2025 (Postdoc)), ETRI AI Safety Institute (Researcher)
3. **Hyunsoo Lee** (03/2021–02/2026 (PhD)), LG Electronics (Researcher)
4. **Gyu Seon Kim** (03/2023–02/2026 (PhD), 03/2026–04/2026 (Postdoc)), RealtimeVisual (Researcher, *Military Service Exception*)
5. **Seok Bin Son** (03/2022–02/2027 (PhD), 03/2027–Present (Postdoc)), Korea University (Postdoctoral Scholar)

■ M.S. Alumni

1. **Kyeongseon Kim** (09/2017–08/2019), KAIST (Ph.D. Student in Computer Science)
2. **Dohyun Kwon** (03/2018–02/2020), Hyundai Marine & Fire Insurance (Researcher)
3. **Dohyun Kim** (03/2018–02/2020), Naver Webtoon (Researcher)
4. **MyungJae Shin** (03/2018–02/2020), Naver (Researcher)

5. **Jaeho Choi** (03/2019–02/2021), Korea Meteorological Administration (Researcher, *Military Service Exception*)
6. **Yoo Jeong (Anna) Ha** (03/2021–02/2023), The University of Chicago (Ph.D. Student in Computer Science)
7. **Jaehyun Chung** (09/2023–08/2025), Hanwha Aerospace (Researcher)
8. **Chaemoon Im** (03/2024–02/2026), Korea University Net-Zero CAFE Research Center (Researcher)
9. **Yeryeong Cho** (03/2024–02/2026), Korea University (Ph.D. Student in Electrical and Computer Engineering)
10. **Hyojun Ahn** (03/2025–02/2027), Korea University (Ph.D. Student in Electrical and Computer Engineering)

■ Former Research Professors and Postdoctoral Scholars

1. **Minseok Choi** (09/2018–02/2019, jointly with Prof. Andreas F. Molisch (USC)), **Kyung Hee University (Professor)**
2. **Soyi Jung** (03/2021–08/2021, jointly with Prof. Marco Levorato (UC-Irvine)), **Ajou University (Professor)**
3. **Ju-Hyung Lee** (03/2022–02/2023, jointly with Prof. Andreas F. Molisch (USC)), Nokia USA (Researcher)

■ Former Full-Time Researchers

- **Rhoan Lee**, LG CNS (Researcher) (M.S. in Statistics, Columbia University in the City of New York)
- **Daeyeun Kim** (07/2025–07/2026), Korea University (Ph.D. Student in Electrical and Computer Engineering)
- **Donggun Lee** (04/2026–08/2026), The University of Michigan (B.S. Student in Data Science)

IEEE Activities (Membership, Editorial Boards, and Services)

- *Senior Member (2018–)*, *Member (2006–2017)*
- *Editor (2026–)*, **IEEE Transactions on Machine Learning in Communications and Networking** (Area 1: Learning for PHY/MAC Communications & Signal Processing)
- *Editor (2026–)*, **Quantum Machine Intelligence (Springer-Nature)**
- *Associate Editor (2025–)*, **ACM Computing Surveys**
- *Associate Editor (2025–)*, **IEEE Communications Surveys and Tutorials**
- *Editor (2023–)*, **IEEE Internet of Things Journal**
- *Associate Editor (2020–)*, **IEEE Transactions on Vehicular Technology** (Area: Vehicular Electronics and Systems)
- *Guest Editor*, **Journal of Communications and Networks** (S.I. on Quantum Technologies for Communication Systems)
- *Guest Editor*, **IEEE Communications Standards Magazine** (S.I. on Recent and Future Evolution of Wi-Fi)

References

- **Prof. Andreas F. Molisch *(Fellow of the IEEE)***, *Ph.D. Research and Dissertation Advisor*
  – Professor of Electrical and Computer Engineering at the University of Southern California (Los Angeles, CA, USA)
  – E-mail: molisch@usc.edu